

Bernoulli NML Selects Relative-Periodic Domains

A fitted relative-periodic scaffold isolates defect masks that retain visible structure in several 2D cellular automata

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Affiliation to be added

Main result. A single Bernoulli-NML selector fits a global relative-periodic scaffold to finite spacetimes in 1D, 2D, and 3D cellular automata. In the strongest 2D cases, the remaining defect mask is sparse, but it does not collapse to featureless noise.

Problem

Cellular automata often contain broad domains together with particles, interfaces, and other persistent defects. For a finite spacetime, the practical question is which global period and shift give the best explanation of the observed pattern.

Guiding idea. First fit the global scaffold, and then inspect the residual mask that remains.

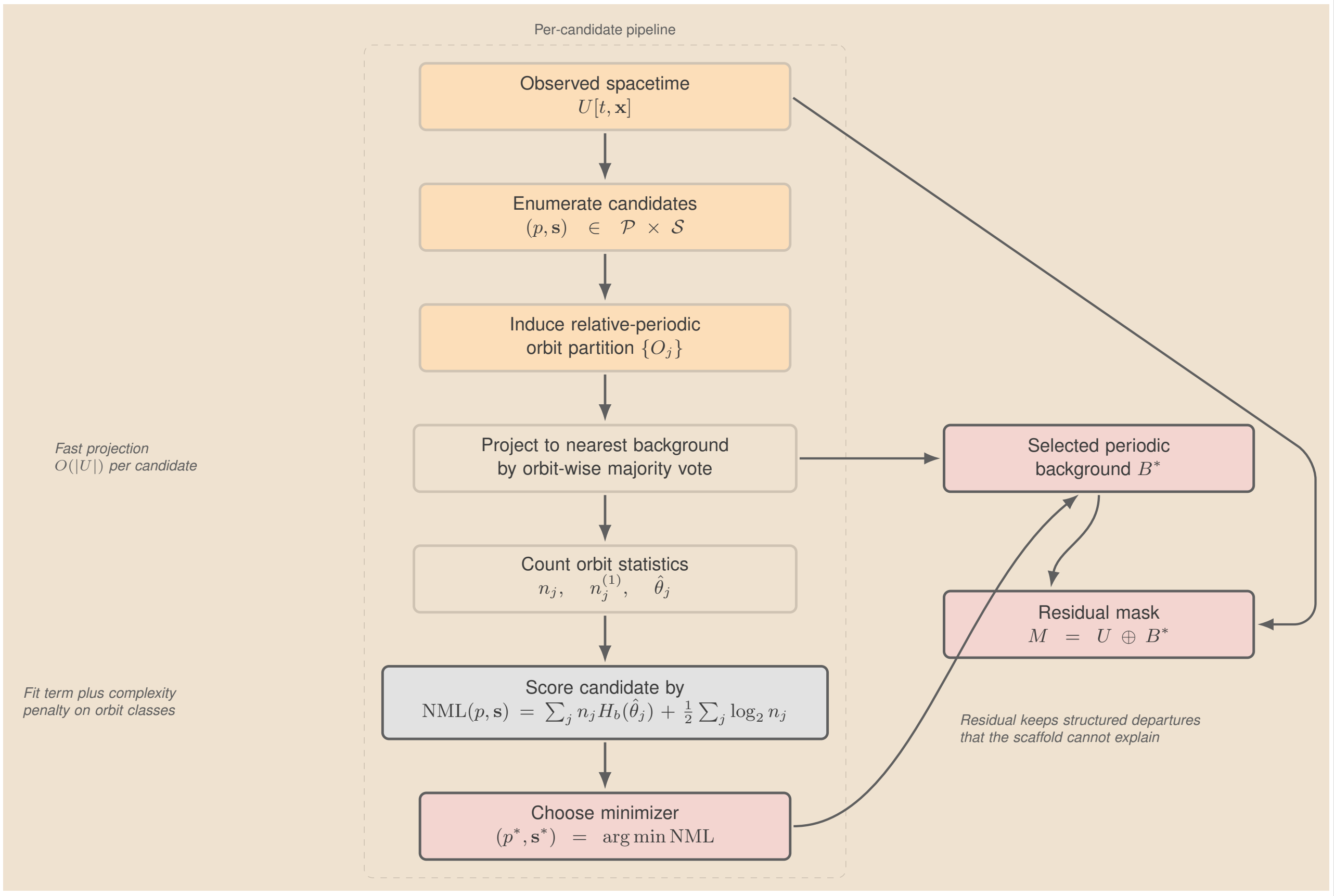
Selector

- A candidate (p, s) defines a relative-periodic model class and its orbit partition.
- Majority vote on each orbit yields the fitted background $B^*(p, s)$.
- The residual mask is $M = U \oplus B^*(p, s)$, and Bernoulli NML ranks the candidates.

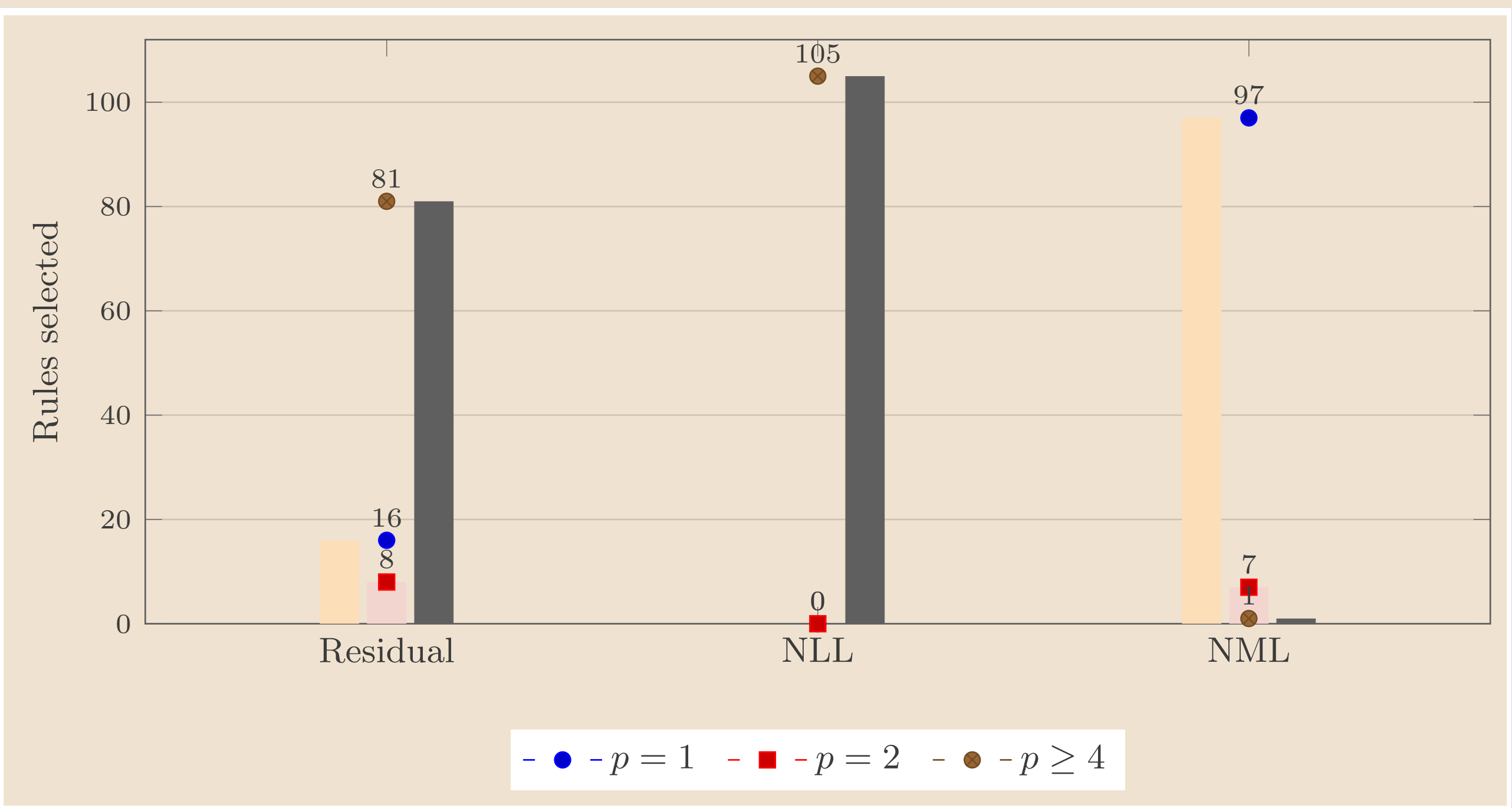
$$\mathcal{B}(p, s) = \{B : B[t + p, \mathbf{x} + s \bmod \mathbf{D}] = B[t, \mathbf{x}]\}$$

$$\text{NML}(p, s) = \sum_j n_j H_b(\hat{\theta}_j) + \frac{1}{2} \sum_j \log_2 n_j$$

Here n_j is the size of orbit class j and $\hat{\theta}_j$ is its fitted Bernoulli parameter. Pure orbit classes lower the fit term, but extra template bits raise the penalty.



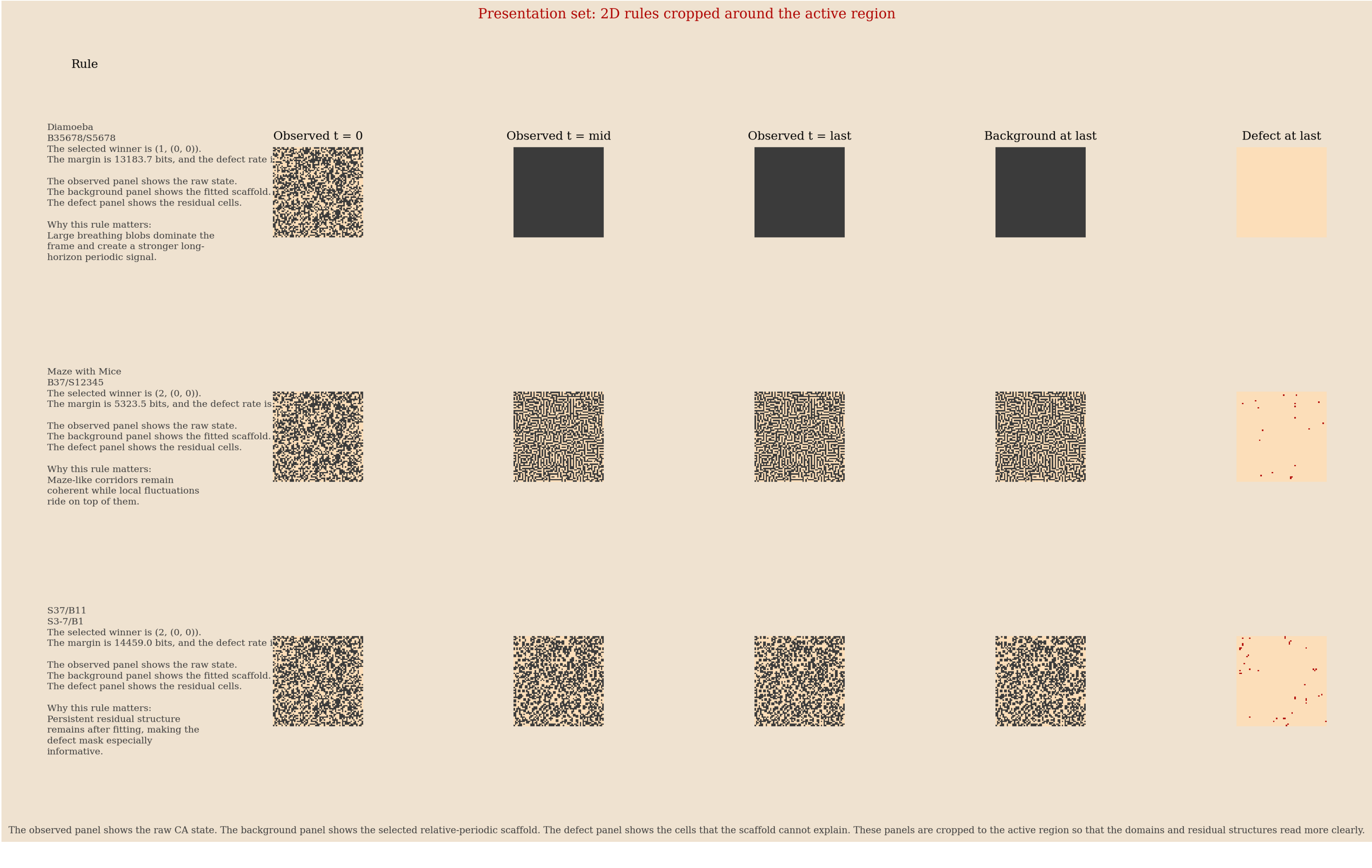
Selector Ablation



Residual minimization and Bernoulli NLL both inflate period because fit can only improve along refinement chains. The Bernoulli NML penalty is what keeps the selector parsimonious.

2D Main Result

These panels distinguish the broad scaffold from the residual structure that remains after subtraction.

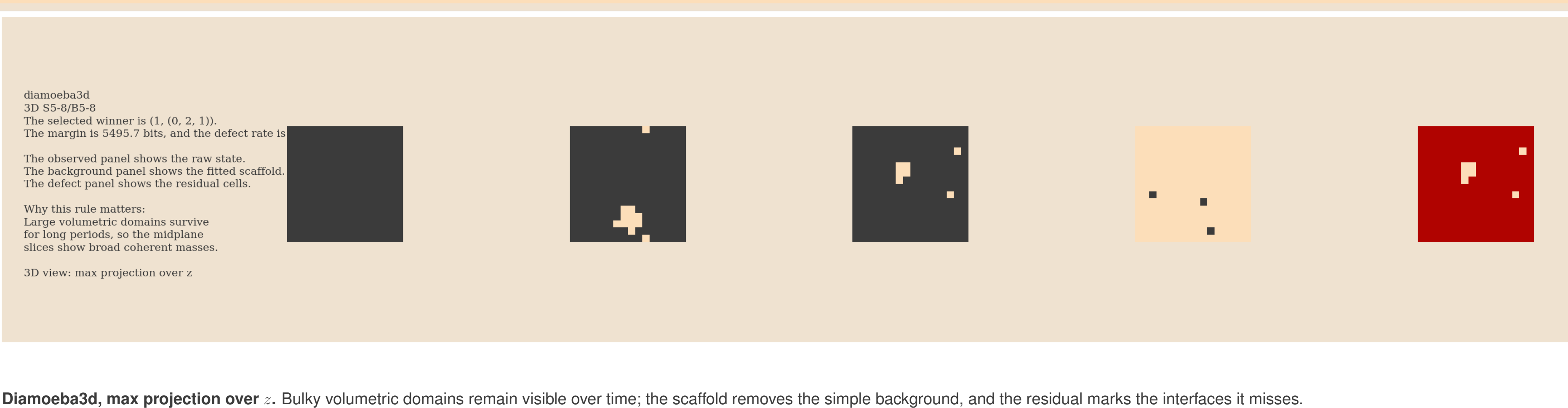


Main visual claim. In 2D, the fitted scaffold explains broad regular domains, but the residual mask does not become featureless. Instead, it often retains sparse coherent structure.

Focal 2D Cases

- In **Diaemoeba**, large breathing domains lead the selector to choose $(p, s) = (1, (0, 0))$, with defect rate 0.077.
- In **Maze with Mice**, a coherent maze background yields winner $(2, (0, 0))$, with defect rate 0.008.
- In **S37/B11**, a low-defect background coexists with persistent residual structure; the presentation run gives winner $(2, (0, 0))$ and defect rate 0.015.

3D Example



Diaemoeba3d, max projection over z. Bulky volumetric domains remain visible over time; the scaffold removes the simple background, and the residual marks the interfaces it misses.

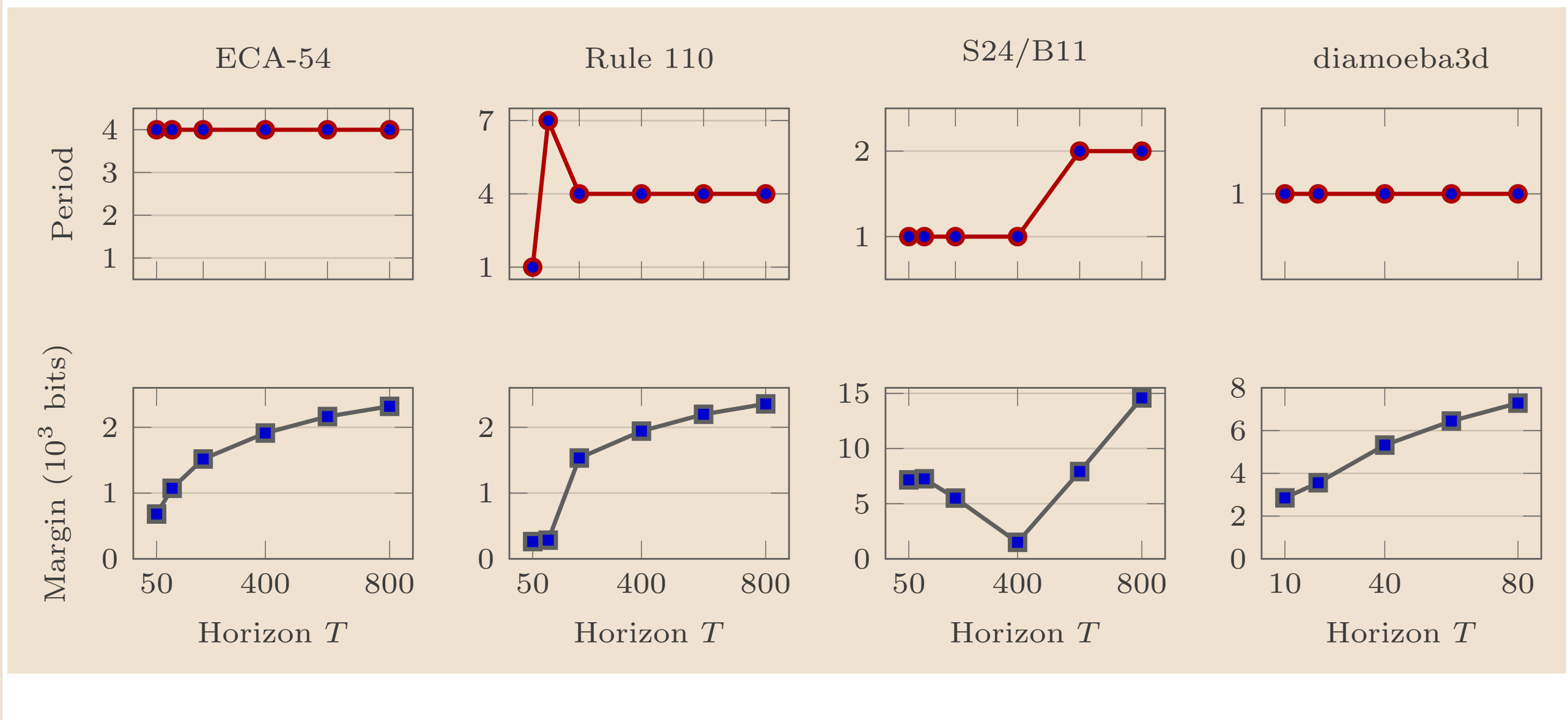
Rule Mechanisms



This manuscript figure summarizes the local rule families behind the poster examples: 1D lookup tables, 2D Moore-neighborhood birth/survival counts, and 3D totalistic thresholds.

Stability Across Horizon

As the observation horizon grows, the selected period usually stabilizes quickly.

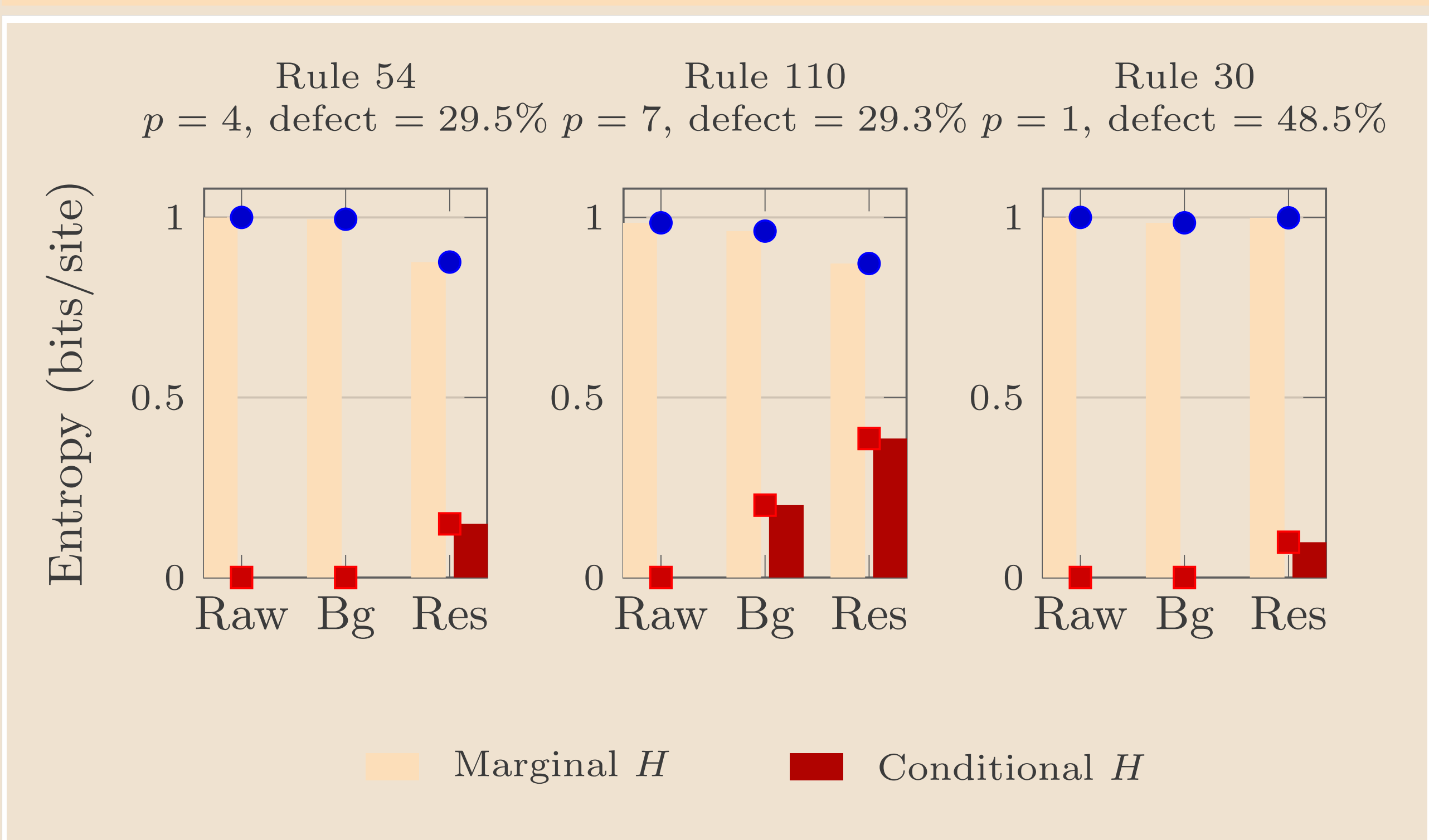


$$\text{NML}(c, T) = \lambda_c N(T) + \frac{k_c}{2} \log_2 T + o(N(T))$$

When candidates have distinct asymptotic rates, the worse rate is eventually excluded and the winner locks.

- ECA-54** locks immediately to period 4.
- ECA-110** passes through a short transient and then locks to period 4 with shift -2 .
- S24/B11** changes later, because longer horizons reveal additional periodic structure.
- Diaemoeba3d** corrects a small-sample transient and then locks to period 1.

Residual Entropy



The manuscript entropy decomposition makes the poster's qualitative point more precise: the fitted periodic background keeps most predictability, while the residual has substantially lower conditional entropy than marginal entropy. The mask is sparse, but it is not white noise.

Takeaways

- One Bernoulli-NML selector provides usable periodic scaffolds in 1D, 2D, and 3D cellular automata.
- The complexity penalty changes the selected period qualitatively; it is not a cosmetic adjustment.
- In 2D, residual masks expose structured defects and therefore make broader surveys feasible.
- The selector is a front end for richer local analysis, not a replacement for computational mechanics.

Repository and figure sources. The selector diagram, selector-ablation chart, and stabilization chart are generated from the TikZ sources in poster/figures/. The presentation panels are generated in outputs/alife_2026/rule_diagrams. The local-rule mechanisms panel is reused from paper/figures/alife_rule_mechanisms.png.

Supporting data. Control summaries are in outputs/alife_2026/null_controls. Multi-seed and scaling summaries are in outputs/strengthening_v2. The entropy comparison is adapted from the Complex Systems manuscript bundle.

Project link. https://github.com/zitterbewegung/relative_symmetry_repair



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